NIOS Class 10 (211) Mathematics — Top 50 Most Repeated Objective Questions (MCQs, Blanks & True/False)

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

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Target: Master the 42 Objective Marks in Section A

> **Why this file is your biggest asset:**
> In NIOS Mathematics 211, **Section A carries 42 marks out of 85** (nearly 50% of the entire paper!).
> If you score 35+ on these quick, 1-step objective questions, you have already guaranteed a passing score before even opening Section B!

Part 1: Multiple Choice Questions (1 Mark Each)

1. Quadratic Equations & Algebra

Q1. The discriminant of the quadratic equation $x^2 - 4x + 3 = 0$ is:
(A) 4
(B) -4
(C) 2
(D) 0

> **Answer: (A) 4**
> *Explanation*: $D = b^2 - 4ac = (-4)^2 - 4(1)(3) = 16 - 12 = 4$. Since $D > 0$, roots are real and distinct.

Q2. If the quadratic equation $kx^2 - 4x + 1 = 0$ has two equal real roots, then the value of k is:

(A) 2

(B) 4

(C) -4

(D) 16

> **Answer: (B) 4**
> *Explanation*: For equal roots, $D = b^2 - 4ac = 0 \implies (-4)^2 - 4(k)(1) = 0 \implies 16 = 4k \implies k = 4$.

Q3. The roots of the quadratic equation $x^2 - 25 = 0$ are:

(A) 5

(B) -5

(C) ± 5

(D) 25

> **Answer: (C) ± 5**
> *Explanation*: $x^2 = 25 \implies x = \pm\sqrt{25} = \pm 5$.

Q4. The 10th term of the AP $2, 7, 12, 17, \ldots$ is:

(A) 45

(B) 47

(C) 50

(D) 52

> **Answer: (B) 47**
> *Explanation*: $a = 2, d = 7 - 2 = 5$. Formula: $a_n = a + (n-1)d \implies a_{10} = 2 + (10-1) \times 5 = 2 + 45 = 47$.

Q5. In an AP, if $a = 5, d = 3$, and $a_n = 50$, then n is:

(A) 14

(B) 15

(C) 16

(D) 17

Answer: (C) 16
Explanation: $50 = 5 + (n-1) \times 3 \implies 45 = 3(n-1) \implies n - 1 = 15 \implies n = 16.$

Q6. The sum of first 10 terms of the AP $1, 3, 5, 7, \ldots$ is:
(A) 100
(B) 90
(C) 110
(D) 50

Answer: (A) 100
Explanation: The sum of the first n odd natural numbers is $n^2 \implies 10^2 = 100.$

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2. Geometry & Coordinate Geometry

Q7. The distance between the points (a, b) and $(-a, -b)$ is:
(A) $\sqrt{a^2 + b^2}$
(B) $2\sqrt{a^2 + b^2}$
(C) $a + b$
(D) $2(a + b)$

Answer: (B) $2\sqrt{a^2 + b^2}$
Explanation: $d = \sqrt{(-a - a)^2 + (-b - b)^2} = \sqrt{(-2a)^2 + (-2b)^2} = \sqrt{4a^2 + 4b^2} = 2\sqrt{a^2 + b^2}.$

Q8. The mid-point of the line segment joining the points $(-1, 3)$ and $(3, 5)$ is:
(A) $(1, 4)$
(B) $(2, 4)$
(C) $(1, 8)$
(D) $(2, 8)$

Answer: (A) (1, 4)
Explanation: $M = \left(\frac{-1+3}{2}, \frac{3+5}{2}\right) = \left(\frac{2}{2}, \frac{8}{2}\right) = (1, 4).$

Q9. The distance of the point $P(3, 4)$ from the origin $(0, 0)$ is:
(A) 3

(B) 4
(C) 5
(D) 7

> **Answer: (C) 5**
> *Explanation*: $d = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$

Q10. The length of tangent from an external point A to a circle of radius $7\ \text{cm}$ is $24\ \text{cm}$. The distance of point A from the centre is:
(A) $25\ \text{cm}$
(B) $31\ \text{cm}$
(C) $17\ \text{cm}$
(D) $26\ \text{cm}$

> **Answer: (A) 25 cm**
> *Explanation*: By Pythagoras theorem, $d = \sqrt{r^2 + \text{tangent}^2} = \sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25\ \text{cm}.$

Q11. The angle subtended by a diameter at any point on the semicircle is:
(A) 45°
(B) 60°
(C) 90°
(D) 180°

> **Answer: (C) 90°**
> *Explanation*: Angle in a semicircle is always a right angle (90°).

Q12. In a cyclic quadrilateral $ABCD$, if $\angle A = 70^\circ$, then the measure of $\angle C$ is:
(A) 70°
(B) 110°
(C) 90°
(D) 120°

> **Answer: (B) 110°**
> *Explanation*: Opposite angles of a cyclic quadrilateral sum to $180^\circ \implies \angle C = 180^\circ - 70^\circ = 110^\circ.$

3. Trigonometry

Q13. The value of $\sec 60^\circ$ is:
(A) $\frac{1}{2}$
(B) $\frac{\sqrt{3}}{2}$
(C) $\sqrt{2}$
(D) 2

> **Answer: (D) 2**
> *Explanation*: $\cos 60^\circ = \frac{1}{2} \implies \sec 60^\circ = \frac{1}{\cos 60^\circ} = 2.$

Q14. The value of $\sin^2 30^\circ + \cos^2 30^\circ$ is:
(A) 0
(B) 1
(C) $\frac{1}{2}$
(D) 2

> **Answer: (B) 1**
> *Explanation*: Identity $\sin^2 \theta + \cos^2 \theta = 1$ holds for all angles.

Q15. If $\tan \theta = 1$, then the value of θ is:
(A) 30°
(B) 45°
(C) 60°
(D) 90°

> **Answer: (B) 45°**
> *Explanation*: $\tan 45^\circ = 1.$

Q16. The value of $\tan 60^\circ \cdot \cot 60^\circ$ is:
(A) $\sqrt{3}$
(B) $\frac{1}{\sqrt{3}}$
(C) 1
(D) 0

Answer: (C) 1
Explanation: $\cot\theta = \frac{1}{\tan\theta} \implies \tan\theta \cdot \cot\theta = 1.$

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4. Mensuration

Q17. Area of a square whose perimeter is $44\ \text{cm}$ is:
(A) $121\ \text{cm}^2$
(B) $144\ \text{cm}^2$
(C) $44\ \text{cm}^2$
(D) $169\ \text{cm}^2$

Answer: (A) 121 cm²
Explanation: $\text{Perimeter} = 4a = 44 \implies a = 11\ \text{cm} \implies \text{Area} = a^2 = 11^2 = 121\ \text{cm}^2.$

Q18. The volume of a cylinder of the same base radius and height as that of a cone is:
(A) Equal to volume of cone
(B) 2 times the volume of cone
(C) 3 times the volume of cone
(D) $\frac{1}{3}$ the volume of cone

Answer: (C) 3 times the volume of cone
Explanation: $V_{\text{cylinder}} = \pi r^2 h = 3 \times \left(\frac{1}{3}\pi r^2 h\right) = 3 \times V_{\text{cone}}.$

Q19. Length, breadth, and height of a cuboid are $3\ \text{cm}, 4\ \text{cm}$, and $5\ \text{cm}$ respectively. Its volume is:
(A) $12\ \text{cm}^3$
(B) $60\ \text{cm}^3$
(C) $20\ \text{cm}^3$
(D) $48\ \text{cm}^3$

Answer: (B) 60 cm³
Explanation: $V = l \times b \times h = 3 \times 4 \times 5 = 60\ \text{cm}^3.$

Q20. If the radius of a sphere is doubled, its surface area becomes:
(A) 2 times
(B) 3 times

(C) 4 times
(D) 8 times

> **Answer: (C) 4 times**
> *Explanation*: Surface Area $= 4\pi r^2$. When $r \to 2r$, $\text{New Area} = 4\pi(2r)^2 = 4 \times (4\pi r^2)$.

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5. Statistics & Probability

Q21. The middle-most observation of a data set arranged in ascending or descending order is called:
(A) Mean
(B) Median
(C) Mode
(D) Range

> **Answer: (B) Median**

Q22. If Mean $= 10$ and Median $= 12$, then Mode using the empirical formula is:
(A) 14
(B) 16
(C) 18
(D) 20

> **Answer: (B) 16**
> *Explanation*: $\text{Mode} = 3\,\text{Median} - 2\,\text{Mean} = 3(12) - 2(10) = 36 - 20 = 16$.

Q23. If $P(E) = 0.05$, then the probability of 'not E' is:
(A) 0.95
(B) 0.05
(C) 1.05
(D) 0

> **Answer: (A) 0.95**
> *Explanation*: $P(\bar{E}) = 1 - P(E) = 1 - 0.05 = 0.95$.

Q24. If $P(E) + P(\bar{E}) = y$, then the value of $y^3 - 4$ is:
(A) 4

(B) 3
(C) -3
(D) 0

> **Answer: (C) -3**
> *Explanation*: Since $P(E) + P(\bar{E}) = 1 \implies y = 1 \implies y^3 - 4 = 1^3 - 4 = -3$. *(Very popular NIOS question!)*

Q25. Two coins are tossed simultaneously. The probability of getting at least one head is:
(A) $\frac{1}{4}$
(B) $\frac{1}{2}$
(C) $\frac{3}{4}$
(D) 1

> **Answer: (C) 3/4**
> *Explanation*: Sample space $= \{HH, HT, TH, TT\}$ (4 outcomes). Favourable $= \{HH, HT, TH\}$ (3 outcomes) $\implies \frac{3}{4}$.

Q26. Two dice are thrown together. The probability that they show different numbers is:
(A) $\frac{1}{6}$
(B) $\frac{5}{6}$
(C) $\frac{1}{36}$
(D) $\frac{35}{36}$

> **Answer: (B) 5/6**
> *Explanation*: Total outcomes $= 36$. Same numbers (doublets) $= \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\} = 6 \implies P(\text{same}) = \frac{6}{36} = \frac{1}{6}$. Therefore, $P(\text{different}) = 1 - \frac{1}{6} = \frac{5}{6}$.

Q27. A card is drawn from a well-shuffled deck of 52 cards. The probability that it is an Ace is:
(A) $\frac{1}{52}$
(B) $\frac{1}{13}$
(C) $\frac{4}{13}$
(D) $\frac{1}{4}$

Answer: (B) 1/13
Explanation: Total Aces $= 4 \implies P = \frac{4}{52} = \frac{1}{13}$.

Part 2: Fill in the Blanks & Match Questions (2 Marks Each / 1 Mark Per Part)

Q28.
(i) Two factors $(x + 3)$ and _____ are obtained on factorising $x^2 + 8x + 15$.

Answer: $(x + 5)$
Reason: $3 \times 5 = 15$ and $3 + 5 = 8$.

(ii) The value of the polynomial $2x + 3x^2 - 4$ at $x = 0$ is _____.

Answer: -4
Reason: $2(0) + 3(0)^2 - 4 = -4$.

Q29.
(i) Angles in the same segment of a circle are _____.

Answer: **equal**

(ii) If the sum of a pair of opposite angles of a quadrilateral is 180°, the quadrilateral is _____.

Answer: **cyclic**

Q30.
(i) A tangent to a circle intersects it in exactly _____ point(s).

Answer: **one**

(ii) A line intersecting a circle in two points is called a _____.

Answer: **secant**

Q31.

(i) The maximum number of parallel tangents that a circle can have is _____.

> **Answer: two**

(ii) The common point of a tangent to a circle and the circle is called _____.

> **Answer: point of contact**

Q32.

(i) $\sin^2\theta + \cos^2\theta =$ _____.

> **Answer:** 1

(ii) $1 + \tan^2\theta =$ _____.

> **Answer:** $\sec^2\theta$

Q33.

(i) The coordinates of the origin are _____.

> **Answer:** $(0, 0)$

(ii) The y-coordinate of any point on the x-axis is _____.

> **Answer:** 0

Q34.

(i) The probability of a sure (certain) event is _____.

> **Answer:** 1

(ii) The probability of an impossible event is _____.

> **Answer:** 0

Q35.

(i) If cash price of a cycle is ₹2,500 and down payment is ₹500, the unpaid balance is _____.

> **Answer**: **₹2,000** $(2500 - 500)$

(ii) Amount of each installment in 4 equal monthly installments for ₹2,000 (without interest) is _____.

> **Answer**: **₹500** $(2000/4)$

Part 3: True / False Statements (2 Marks Each / 1 Mark Per Part)

Q36.

(i) "Using a ruler and compass it is possible to construct an angle of 25°."

> **Answer**: **FALSE**
> *Reason*: Only multiples of 15° ($15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ \ldots$) or their bisections ($7.5^\circ, 22.5^\circ$) can be constructed. 25° is not constructible.

(ii) "The tangent at any point of a circle is perpendicular to the radius through the point of contact."

> **Answer**: **TRUE**
> *Reason*: This is the fundamental Circle Tangent Theorem.

Q37.

(i) "Two distinct lines can have more than one point in common."

> **Answer**: **FALSE**
> *Reason*: Two distinct straight lines can intersect at at most 1 point.

(ii) "The ratio of the circumference to the diameter of any circle is always equal to π."

> **Answer**: **TRUE**
> *Reason*: $C = \pi d \implies \frac{C}{d} = \pi$.

Q38.

(i) "The probability of an event can be -0.5."

> **Answer**: **FALSE**
> *Reason*: Probability must always satisfy $0 \leq P(E) \leq 1$. It can never be negative.

(ii) "In an AP, the difference between any two consecutive terms is always constant."

> **Answer**: **TRUE**
> *Reason*: That is the definition of common difference d.

Q39.

(i) "The discriminant of the equation $x^2 + 4x + 5 = 0$ is negative."

> **Answer**: **TRUE**
> *Reason*: $D = 4^2 - 4(1)(5) = 16 - 20 = -4 < 0$.

(ii) "The value of $\sin 90^\circ$ is 0."

> **Answer**: **FALSE**
> *Reason*: $\sin 90^\circ = 1$ (while $\cos 90^\circ = 0$).

Q40.

(i) "The curved surface area of a cone is given by $\pi r l$."

> **Answer**: **TRUE**

(ii) "The volume of a sphere of radius r is $\frac{2}{3}\pi r^3$."

> **Answer**: **FALSE**
> *Reason*: Volume of a sphere is $\frac{4}{3}\pi r^3$ (Hemisphere is $\frac{2}{3}\pi r^3$).